Juan Rodrigo Ferro

Software engineer · production GenAI, AWS/GCP architecture, full-stack TypeScript & Python

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Student visa with work rights (48 h/fortnight during session, unrestricted in breaks)

Summary

Software engineer who ships production systems end-to-end: LLM agents with tool-calling on AWS, Next.js applications with their own design systems, and the event-driven cloud architecture underneath. I work in regulated environments where the audit trail, the data-residency question and the cost model are part of the deliverable, and I take engagements from the first client meeting to a signed-off document set. I measure before I conclude.

Experience

Software Engineer, Cloud & GenAI · BWR, software consultancy (Mexico), remote from Sydney — 2025 - present

LLM agent with tool-calling · in production
- Own the AI and serverless layer of a conversational booking agent: Amazon Bedrock Agent, Lambda action group with 4 tools, SQS FIFO, HMAC-signed calls to the product backend.
- Turned a live pilot into 5 shipped fixes; 3 were prompt changes that unit tests cannot cover, which required structured live conversation testing. Specified caller identity from webhook metadata only.

Speech-to-text + LLM cost model · decision memo, system in production
- Showed transcription is 80-99.9% of pipeline cost and recommended migrating it, not the LLM; modelled 6×-36× savings with break-even by volume, and gated the migration on a pilot because of Whisper's silence-hallucination risk.

Event-driven integration with immutable audit · competitive tender, outcome pending
- Owned the AWS design: mTLS ingress, format translation on Lambda, SQS FIFO, EventBridge, S3 Object Lock with hash chaining. 7 decisions with alternatives, 65 verified sources; committed a measured p95 SLO instead of an unbacked guarantee. Gap analysis found 4 disqualifying risks and 17 missing questions.

Document AI with a human in the loop, federal agency · requirements and architecture
- Took requirements from a client meeting to a v2.0 validated by a legal specialist, with evidence marking on every requirement; designed delivery waves where the first uses no AI as the permanent degraded path; 3 architecture variants, 3 ADRs, 15 technical decisions.

Next.js web application · in development, deployed to QA
- Built all 5 screens, the design system and Auth.js v5 authentication with lockout; led a security audit remediation, 20 of 24 findings closed (CSP nonces, rate limits, append-only audit via triggers).
- Diagnosed a production data bug by measurement: 13 rows to 5, payload 1,331 KB to 461 KB. Codebase at 842 tests green.

Other platform work
- Backward-compatible API reorganisation in a multi-tenant PHP/Angular platform, keeping old routes alive for installed mobile clients; React Native features shipped with OTA release sequencing; data-platform architecture for 130 TB with a residency assumption invalidated by verification; found a seven-figure double-count in a commercial estimate before it reached the client.

Independent projects

Factotum: infrastructure for AI coding agents · open source, daily use
- Launch, follow and approve Claude Code agents from a phone PWA on your own hosts: modular core, headless CLI, Web Push approvals, append-only session logs. 1,607 tests, 16 ADRs.

Book-collection RAG chatbot · volunteer, delivered
- Qdrant + reranking for a community group of about 300 people; evaluated with deliberately confusable sources, then with 9 administrators.

Education

Master of Information Systems Management · Macquarie University, Sydney — 2026 - 2027 (in progress)

Security management (ISO 27001/27005, NIST CSF), project and risk management, enterprise strategy, IS design.

B.Eng. Computer Technology (Ingeniería en Tecnologías Computacionales) · Tec de Monterrey — 2023 - 2027 (in progress)

Certifications

AWS Certified Generative AI Developer, Professional · Oracle OCI 2025 Generative AI Professional · AWS Certified AI Practitioner · Oracle OCI 2025 AI Foundations Associate · AWS Certified Cloud Practitioner

Skills

Cloud	AWS (Lambda, API Gateway, SQS, EventBridge, S3 Object Lock, KMS, Bedrock, Transcribe), GCP (Vertex AI, BigQuery, Cloud Run), Oracle Cloud
AI	LLM agents and tool-calling, RAG, prompt regression testing, ASR, model cost and break-even modelling
Engineering	TypeScript, Python, Node.js, PHP, SQL · React, Next.js, React Native · PostgreSQL, Redis · Vitest, Playwright, TDD, ADRs
Languages	Spanish (native), English (full professional), German (limited working)

Awards

DigiEduHack winner, 2025.